RTD IN CSTR

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Objective:
The aim of this experiment is to determine the mean Residence Time Distribution (RTD) in the constant volume stirred tank reactor, to plot exit age distribution curve and to study segregation model, calculating average conversion.

Theory and Background:
In an ideal CSTR, we assume perfect mixing. It is also assumed that the outlet concentration is equal to the tank concentration. In real systems, however, this is rarely the case; there are always non-ideal conditions present, such as channelling and dead zones. The RTD provides a basis for characterising the mixing and flow within the reactors and comparing the behaviour of real reactors with their idealised models. This can aid not only in troubleshooting existing reactors but also in estimating the yield of a given reaction and designing future reactors.

Residence Time Distribution (RTD) of a chemical reactor is a probability distribution function that describes the amount of time fluid elements spend inside a reactor. It characterizes the flow and mixing behavior in chemical reactors, helping evaluate deviations from ideal reactor models such as Continuous Stirred Tank Reactors (CSTRs) or Plug Flow Reactors (PFRs).

The time distributions of different routes that fluid elements take through the reactor may be different due to the fact that different portions of the fluid are in contact with different elements or wall zones of the reactor. The exit age distribution, or residence time distribution, is denoted by E. E is measured in unit of $time^{-1}$**.** The convenient way to represent RTD is in such a way that the area under the curve equals one, i.e. $\int_0^\infty E(t)\, dt = 1$**.** The value E(t)dt is the fraction of the fluid that spends time duration t inside the reactor.

The RTD function E can be found using the equation : $E(t) = \frac{C(t)}{\int_0^\infty C(t)\, dt}$**.**

RTD is used to get the $Mean\ residence\ time\ (t_m)$ and the variance.
$Mean\ residence\ time\ (t_m)$ is the average time fluid elements spend in the reactor It is related to the reactor's volume V and the volumetric flow rate v. Variance is a measure of dispersion around the mean residence time. $\sigma^2\ (variance)\ and\ (t_m)$ help diagnose the performance of real reactors compared to ideal models, assess mixing quality, and predict the reactor's effectiveness in achieving desired conversions.

To do the calculations, we are using the **Segregation model**. In the segregated flow model, the flow through the reactor takes the form of a continuous series of globules. These globules do not exchange material among each other within the fluid for the time taken for their residence in the reaction environment, i.e. they remain segregated. As there is no molecular interchange between globules, each of them acts essentially as a batch reactor.

The mean conversion $(\overline{X})$ in the effluent stream is determined using the average of conversion of all the globules (batch) in the exit stream. $(\overline{X}) = \int_0^\infty X(t)E(t)\,dt$

$X = 1 - e^{-kt}$, k *is the rate constant.*

The integration of all the variables is done using the area under the curve of respective graphs between the respective variables.

The area is calculated using the shoelace formula. The **shoelace formula**, also known as the **surveyors' formula**, is used to find the area of a polygon given its vertices. In our case (of graph), we are considering the area under the curve a polygon. To complete the polygon, in every graph I have added a point (end time, 0).

Shoelace formula:
For a polygon with vertices $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, *the area is given by*:

$$Area = \frac{1}{2} \bullet \left| \sum_{i=1}^{n} \left(x_i y_{i+1} - y_i x_{i+1}\right) \right|$$

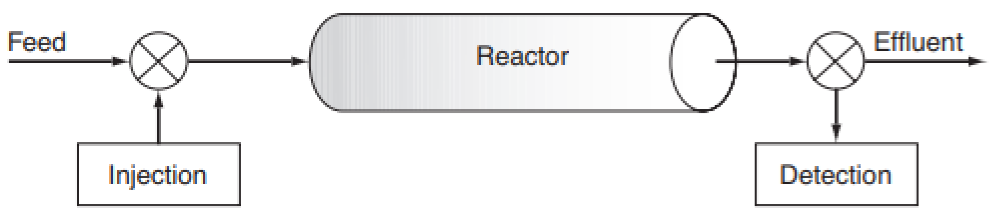

Figure 16-3 Experimental set up to determine E(t).

Figure 0: Schematics of the experiment [1]

<u>Apparatus:</u>

1. A constant inlet flow line
2. Constant volume CSTR
3. Conductivity meter
4. NaOH pellets
5. Magnetic stirrer
6. Syringe
7. 100 ml measuring cylinder

8. Beaker

Procedure:

1. Adjust the inlet flow rate of water to constant value 70 ml/min.
2. Arrange the setup of CSTR using a magnetic stirrer and fix the conductivity cell to the outlet of CSTR.
3. Prepare solution of NaOH dissolving 10gm NaOH in 10 ml distilled water and stir it continually for pulse input.
4. Adjust the stirrer speed so that uniform mixing can be obtained in the reactor.
5. Take the concentrated solution of NaOH in the syringe and inject in the cylinder in one shot at time $t = 0$. Be sure that there is no channeling by injecting it as far away from the outlet as possible.
6. Note down the conductivity values of solution with time at the interval of 0.5 mS drop in the conductivity.
7. Note down the readings until the steady state.
8. Find out the unknown concentration of the exit tracer using concentration vs conductivity plot.
9. Plot concentration vs time.
10. Tabulate $E(t)$, plot exit age distribution curve.

Known parameters:

- $Volume\ of\ the\ reactor\ (V)\ =\ 400\ mL$
- $Volumetric\ flow\ rate\ (v)\ =\ 70\ \left(\frac{mL}{min}\right)$
- $Temperature\ =\ 27\ °C$
- $Rate\ constant\ (k)\ =\ 0.063\ \frac{L}{mol}$

Observation table (mentioned in manual)

Parameter	Value	unit
Temperature, T	27.0000	°C
Rate constant, k	0.0630	l/mol
Reactor volume, V	400.0000	mL

Volumetric flow rate, v	70.0000	ml/min
Space time (τ)	5.7143	min

Inlet concentration (C_{A0}) *is not used here to calculate the conversion* (x).

Here the reaction is 1^{st} *order so there will be no* C_{B0} *term. So* $\Theta_B = \frac{C_{B0}}{C_{A0}}$ *is also not there.*

We are using 10g NaOH in 10 mL water, i.e **25 mM** solution for pulse input.

Formulae:

- $RTD\ function\ E:\ E(t) = \frac{C(t)}{\int_0^\infty C(t)\,dt}$
  - Where $C(t)$ *is the Concentration*
  - t *is the time*
- $Mean\ residence\ time\ t_m = \frac{\int_0^\infty t\,E(t)\,dt}{\int_0^\infty E(t)\,dt}$
- $Variance\ (\sigma^2) = \frac{\int_0^\infty (t-t_m)^2 E(t)\,dt}{\int_0^\infty E(t)\,dt}$
- $Standard\ deviation\ (\sigma) = \sqrt{Variance}$
- $Space\ time\ ,\ \tau = \frac{V}{v} = \frac{400\ mL}{70\ \frac{mL}{min}} = 5.17\ min$
  - *Where* V *is the Volume of the reactor*
  - v *is the Volumetric flow rate*
- $X = 1 - e^{-kt}$
- $Mean\ conversion\ (\overline{X}) = \int_0^\infty X(t)E(t)\,dt$

Readings:

TABLE 1: TABLE CONTAINING EXPERIMENTAL DATA OF CONCENTRATION AND CONDUCTIVITY

Concentration (M)	Conductivity (mS)
0.5	84.27
0.25	39.95

0.125	20.31
0.0625	10.58

TABLE 2: TABLE CONTAINING EXPERIMENTAL DATA OF CONDUCTIVITY AND TIME

Time (s)	Conductivity (mS)
19.46	0
22.46	15
46	30
69.88	45
69.18	60
66.12	75
63.36	90
60.83	105
58.44	120
56.25	135
53.36	150
51.64	165
49.59	180
47.7	195
45.78	210
43.66	225
42.07	240
40.46	255
38.58	270
36.29	285
35.77	300
35.56	315
33.59	330
31.15	345
31.04	360

28.05	375
26.76	390
26.03	405
26.03	420

TABLE 3: TABLE CONTAINING CALCULATED DATA

Conductivity (mS)	Time (s)	Concentration (M)	E (t)	t * E(t)	$\left(t - t_m\right)^2 E(t)$	X	X * E (t)
19.46	0	0.1200	0.0011	0.0000	37.0640	0.0000	0.0000
22.46	15	0.1378	0.0012	0.0186	35.9352	0.6113	0.0008
46	30	0.2769	0.0025	0.0746	60.0722	0.8489	0.0021
69.88	45	0.4180	0.0038	0.1689	74.0326	0.9413	0.0035
69.18	60	0.4139	0.0037	0.2230	58.4778	0.9772	0.0036
66.12	75	0.3958	0.0036	0.2666	43.3475	0.9911	0.0035
63.36	90	0.3795	0.0034	0.3067	31.0376	0.9966	0.0034
60.83	105	0.3645	0.0033	0.3437	21.1790	0.9987	0.0033
58.44	120	0.3504	0.0031	0.3776	13.4733	0.9995	0.0031
56.25	135	0.3375	0.0030	0.4091	7.7086	0.9998	0.0030
53.36	150	0.3204	0.0029	0.4316	3.6126	0.9999	0.0029
51.64	165	0.3102	0.0028	0.4596	1.1634	1.0000	0.0028
49.59	180	0.2981	0.0027	0.4818	0.0791	1.0000	0.0027
47.7	195	0.2869	0.0026	0.5024	0.2357	1.0000	0.0026
45.78	210	0.2756	0.0025	0.5197	1.4933	1.0000	0.0025
43.66	225	0.2631	0.0024	0.5315	3.6976	1.0000	0.0024
42.07	240	0.2537	0.0023	0.5467	6.7817	1.0000	0.0023
40.46	255	0.2441	0.0022	0.5591	10.6094	1.0000	0.0022
38.58	270	0.2330	0.0021	0.5650	14.9645	1.0000	0.0021
36.29	285	0.2195	0.0020	0.5618	19.5393	1.0000	0.0020
35.77	300	0.2164	0.0019	0.5830	25.5081	1.0000	0.0019
35.56	315	0.2152	0.0019	0.6087	32.4378	1.0000	0.0019

33.59	330	0.2035	0.0018	0.6032	38.1984	1.0000	0.0018
31.15	345	0.1891	0.0017	0.5859	43.2395	1.0000	0.0017
31.04	360	0.1885	0.0017	0.6093	51.5733	1.0000	0.0017
28.05	375	0.1708	0.0015	0.5752	55.1150	1.0000	0.0015
26.76	390	0.1632	0.0015	0.5715	61.3175	1.0000	0.0015
26.03	405	0.1589	0.0014	0.5778	68.7751	1.0000	0.0014
26.03	420	0.1589	0.0014	0.5992	78.4894	1.0000	0.0014

Results and discussion:

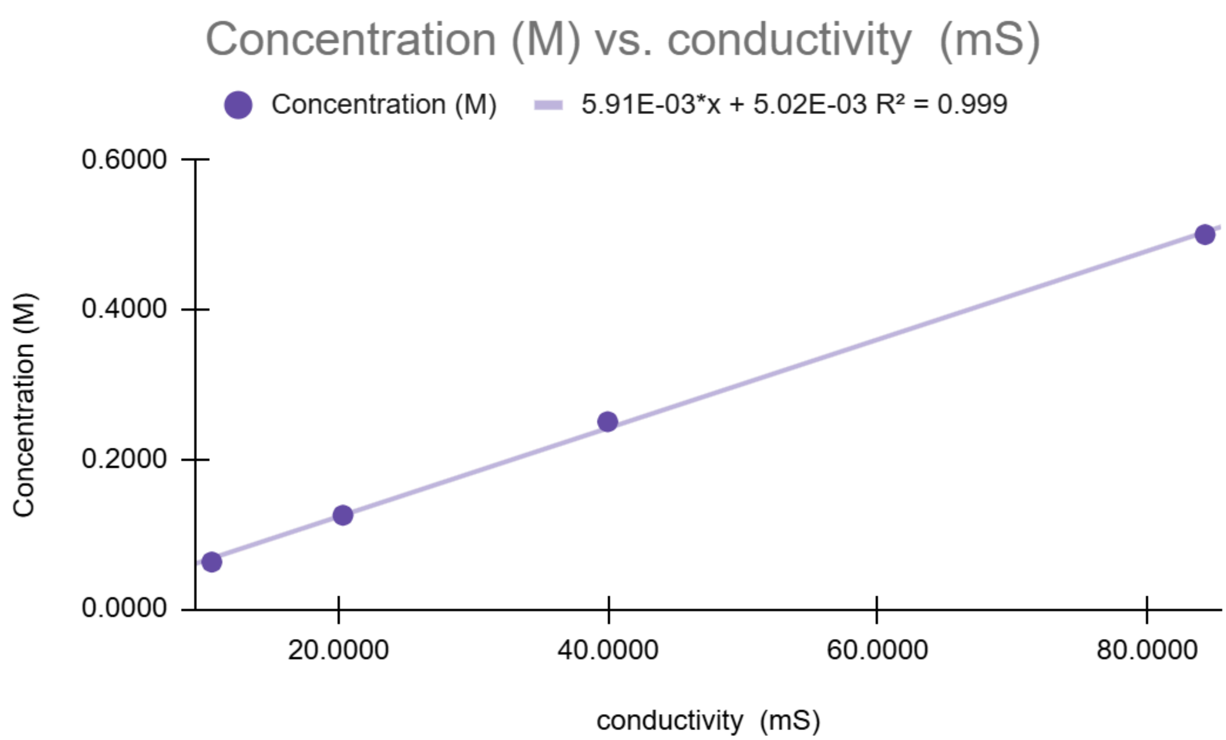

Figure 1: Figure showing the between Concentration (M) vs Conductivity (mS)

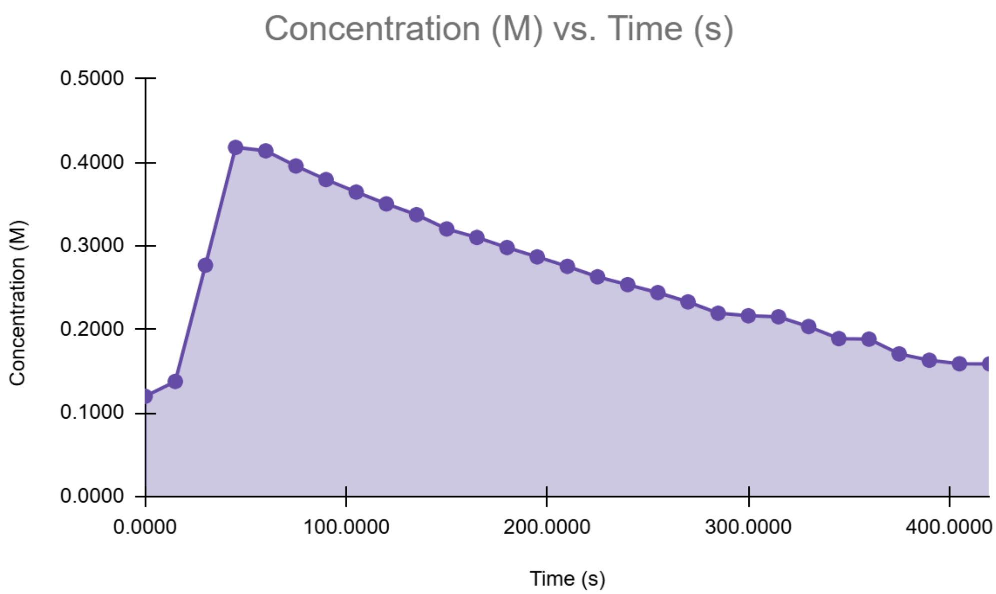

Figure 2: Figure showing the plot between Concentration (M) vs Time (s)

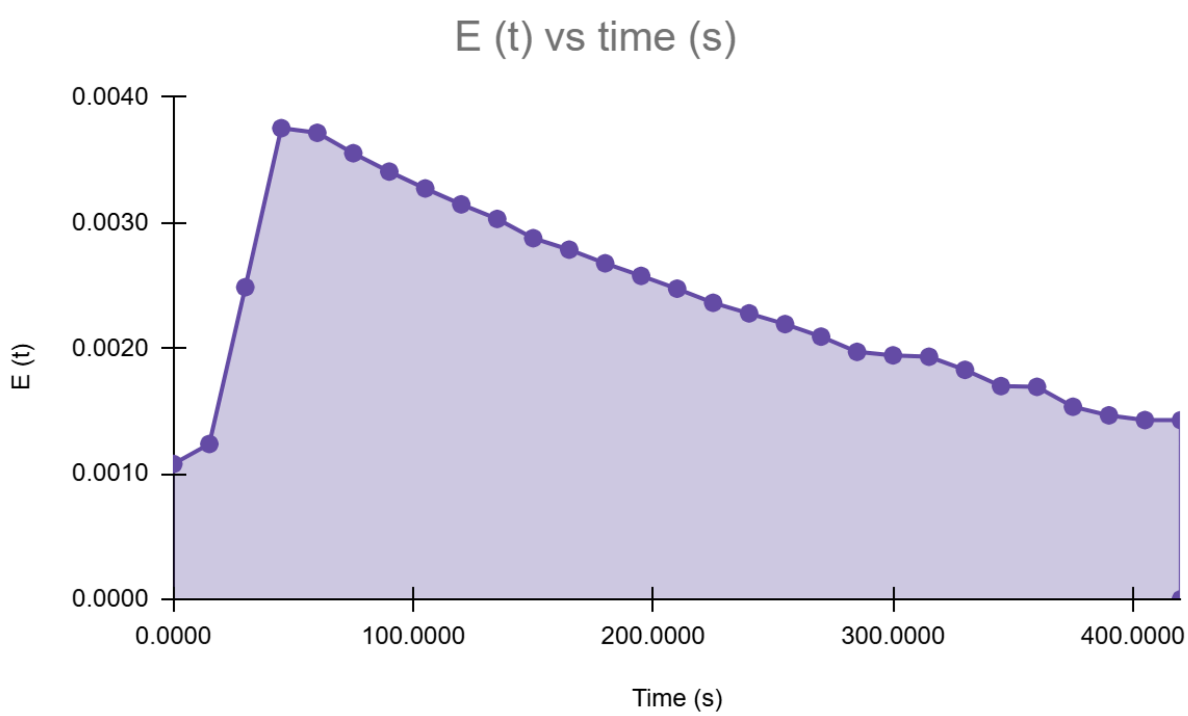

Figure 3: Figure showing the plot between E(t) vs Time (s)

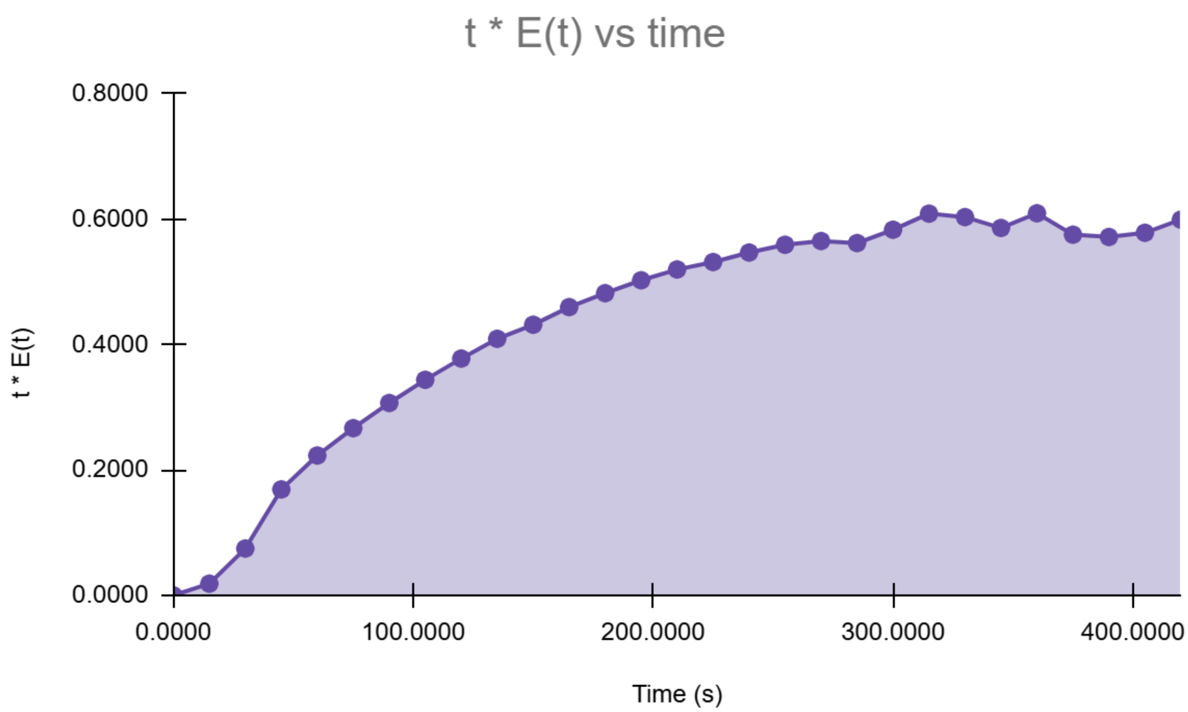

*Figure 4: Figure showing the plot between t * E(t) vs Time (s)*

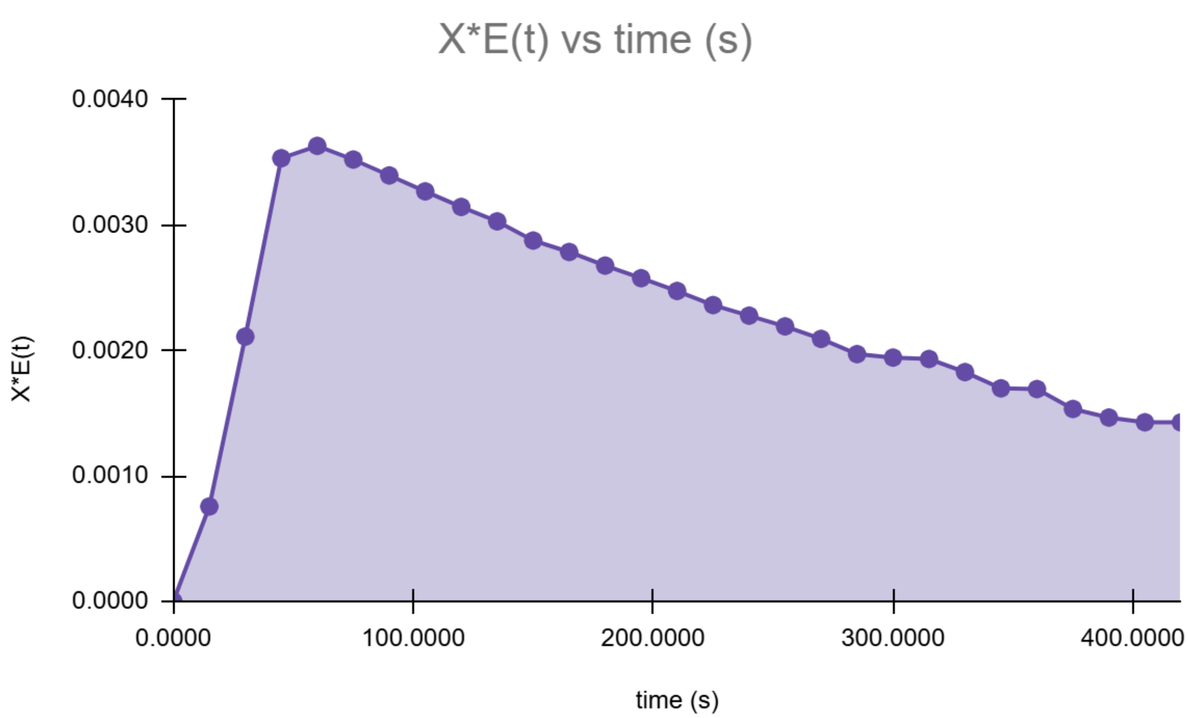

*Figure 5: Figure showing the plot between X * E(t) vs Time (s)*

The mean residence time for the CSTR is = 185.43 s
Standard deviation for the CSTR is = 112.34
Mean conversion using segregation model = 0.97

From figure 1, we can see that there is a linear relationship present between Concentration and conductivity, i.e more is the concentration, more is the conductivity.

Figure 2 shows the relationship between concentration and time. We can see that there is an increase in the concentration initially as we have injected the NaOH (pulse input), the NaOH starts mixing with the reactor contents. Due to the flow and mixing happening in the reactor, with time, the NaOH concentration at the exit is bound to rise. The concentration then becomes a maximum when most of the NaOH injected starts exiting the reactor. This equals the mean residence time (t_m) of the average fluid elements that spent typical amounts of time in the reactor. After maximum concentration, the NaOH concentration decreases, as most of the NaOH has already exited, and the remaining portions undergo dilution due to mixing and less residual NaOH amounts.

Figure 3 and figure 5 shows the relationship between E(t) (Exit age distribution) and Time and X*E(t) vs time respectively. The nature of both the curves are the same as the concentration-time curve. E(t) is a probability density function derived from the concentration data. It is normalized so that the area under the curve equals 1. Therefore, the E(t) vs. time has the same shape as the concentration vs. time curve, but its scale is different due to normalization. Conversion X(t) depends on the time fluid spends in the reactor and the kinetics of the reaction. The product X*E(t) represents the fraction of the NaOH exiting with both a specific age and conversion. Since E(t) has the same shape as concentration, and X(t) generally increases with time, the curve for X · E(t) vs. time also follows a similar pattern as E(t).

Figure 4 shows the t . E(t) vs t curve and it is an increasing curve. The curve represents the weighted residence time distribution, which highlights the contribution of different residence times to the overall mean residence time $\left(t_m\right)$.

The final results:

Mean residence time (t_m)	185.43 s
Variance (σ^2)	12621.447
Standard deviation (σ)	112.3452
mean conversion ($\overline{X}$)	0.97

Conclusion:

The experiment successfully determined the mean Residence Time Distribution (RTD) for a constant volume CSTR. By injecting a pulse input of 25 milliMolar NaOH and analyzing its concentration over time at the reactor outlet, the exit age distribution (E(t)) was plotted, and the mean residence time $\left(t_m\right)$ was calculated. The experiment demonstrated that the feed introduced into the CSTR at any moment becomes completely mixed with the existing reactor contents. Some fluid elements exit almost immediately due to continuous withdrawal, while others remain longer, resulting in a distribution of residence times. The shape and spread of the (E(t)) curve provided insights into the reactor's mixing efficiency and deviations from ideal behavior.

The calculated Mean residence time (t_m) is 185.43 s, along with a variance of 12621.447 , a standard deviation of 112.3452, and a mean conversion $(\overline{X}\,)$of 0.97 demonstrating the extent of residence time variations within the reactor.

The segregation model, used in this study, helped account for non-ideal flow by considering the reactor as a series of segregated globules, each undergoing batch reaction independently. The average conversion was calculated, reinforcing the importance of RTD in reactor design and performance evaluation. This analysis highlights how residence time variations influence the overall reaction outcome, emphasizing the critical role of mixing and flow patterns in chemical reactors.

By completing this experiment, I have gained a comprehensive understanding of how RTD affects the performance of the reactor.

Sources of errors:

- While measuring the conductivity there are charges that we may have previous sample liquid in the bulb which is used to measure the conductivity.
- The tip to the conductivity measuring device may have touched the bottom/walls of the beaker.

References:

[1]“6.1 General Considerations.” Available: Link
[2] ‘Lab Manual’ provided in the classroom.
[3]“Art of Problem Solving,” *Artofproblemsolving.com*, 2025. Shoelace_Theorem